Yan Yang

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Machine learning engineer with industry experience in deep learning, generative models, and ML embedding visualization system design. Research background in machine learning, PDE learning, user motivation/affection modeling, information retrieval, and computer vision. M.Sc. in Computer Science (Georgetown University).

SKILLS

ML / AI	PyTorch, GANs, CNNs, backpropagation, PDE learning, PINN, embedding visualization, data augmentation
Languages	Python, C++, JavaScript/TypeScript, GoLang, SQL
Systems	System design, API architecture, database schemas, caching, serialization, Kubernetes, Redis
Research	Reaction-diffusion systems, NLP, information retrieval, PDE modeling, fMRI processing/model, EEG
Tools	Git, Linux, NumPy, React, Node.js, AngularJS, FreeSurfer

EXPERIENCE

Independent Researcher | Self-directed — *Oct 2025 – Present*

- Diagnosed gradient-based inversion for a reaction-diffusion system by backpropagating through unrolled simulation, used as an ablation to analyze PINN components — presented at AI4Physics @ ICML as a poster, with code released on GitHub.
- Re-examined a prior co-authored paper applying Reversal Theory (RT) to search-behavior modeling; identified mismatches between the theory's phenomenological commitments and the paper's statistical operationalizations, and distinguished theory-inspired approximations with faithful implementations — published in *the Journal of Motivation, Emotion, and Personality*.
- Participated in BAMΞ workshop on measurement (online), Models of Consciousness Conference 6 (Sapporo, Japan), and IBRO Autumn School on fMRI (NYU Abu Dhabi).
- One manuscript under review at TMLR; research notes on Zenodo extending prior work on RT, proposing linkages between reversal triggers & predictive allostasis framework / phenomenological state measurement & quantum measurement principles.

Machine Learning R&D Engineer | Deepcell Inc. · Mountain View, CA — *2022 – 2023*

- Led system design of the remodeled neural network embedding visualization system, integrated into the Cluster-Based Sorting product for real-time data exploration during imaging+sorting runs.
- Architecture decisions spanned data storage, database schemas, API specs, serialization/deserialization, caching, parallelization, and incremental development planning.

Research Fellow | KAUST · Thuwal, Saudi Arabia — *2023 – 2024*

- Investigated data-driven methods for learning PDE parameters in reaction-diffusion systems; the underlying math connects to neuroscience modeling, and in this project was applied to diseased leaf image synthesis for agricultural classification.
- Proposed approaches to Arabic speech recognition, e.g., acoustic source separation and spectrogram-based data augmentation.

R&D Intern | Deepcell Inc. · Mountain View, CA — *2021 – 2022*

- Trained GAN models to supplement data for cell image classifiers.
- Led development of a CNN embedding layer visualization tool used company-wide by data and bioinformatics scientists for clustering and correlation analysis.

Member of Technical Staff | Pure Storage · Bellevue, WA — *2020 – 2021*

- Implemented volume snapshot import/restore from company disks into Kubernetes.
- Bug fix of multipath device issues, Python3 migration of bash scripts; led Pure Service Orchestrator v6.0.3's GitHub release.

Research Assistant | Georgetown University – InfoSense Lab · Washington, DC — *2018 – 2019*

- Translated reversal theory into statistical frameworks to model search engine user behavior and improve struggle detection; work published in Discover Computing (Springer Nature, 2024).
- Revamped VR interface for behavioral data collection; provided web support for SIGIR '18 conference demonstration, acknowledged as authorship-worthy by corresponding author.

Earlier experience includes SDE roles at ColorfulClouds Tech, DataMesh, etc. (2013-2018). Full history available on request.

EDUCATION

Ph.D. studies, Computer Science | University of Nevada, Reno · Reno, NV — *2021 – 2022*

M.Sc., Computer Science | Georgetown University · Washington, DC — *2017 – 2019*

B.Eng. | Southeast University · Nanjing, China — *2002 – 2006*

SELECTED PUBLICATIONS & PEER REVIEW

- Yang (2026). Loss Landscape Diagnosis for Gradient-Based Inversion: Disentangling PINN Components. AI4Physics@ICML'26
- Yang (2026). Modeling Limits in Phenomenological Contexts. Journal of Motivation, Emotion, and Personality 14
- Jiyun et al. (2024). Improving Searcher Struggle Detection via the Reversal Theory. Discover Computing 27(51), Springer Nature
- Yang et al. (2022). Two-Step Data Augmentation for Masked Face Detection and Recognition. IMPROVE Conference, SciTePress
- IEEE Transactions on Affective Computing 2025 · IEEE Transactions on Neural Networks and Learning Systems 2024